

Exp. 6(BCS551)-Joins using multiple tables

Objective:By experiment end, participants will master multi-table joins (inner, outer, cross, natural), enabling accurate data combination, relationship querying, and practical result filtering across relational tables.

CROSS JOIN:

Query1:

```
SELECT e.employee_id, e.last_name, e.job_id, e.department_id,d.department_name
FROM hr.employees e
CROSS JOIN hr.departments d
WHERE ROWNUM <= 20;
```

Output1:

	EMPLOYEE_ID	LAST_NAME	JOB_ID	DEPARTMENT_ID	DEPARTMENT_NAME
1	100	King	AD_PRES	90	Administration
2	101	Yang	AD_VP	90	Administration
3	102	Garcia	AD_VP	90	Administration
4	103	James	IT_PROG	60	Administration
5	104	Miller	IT_PROG	60	Administration
6	105	Williams	IT_PROG	60	Administration
7	106	Jackson	IT_PROG	60	Administration
8	107	Nguyen	IT_PROG	60	Administration

NATURAL JOIN:

Query2:

```
SELECT e.employee_id, e.last_name, e.salary, d.department_name
FROM hr.employees e
NATURAL JOIN hr.departments d;
```

Output2:

	EMPLOYEE_ID	LAST_NAME	SALARY	DEPARTMENT_NAME
1	101	Yang	17000	Executive
2	102	Garcia	17000	Executive
3	104	Miller	6000	IT
4	105	Williams	4800	IT
5	106	Jackson	4800	IT
6	107	Nguyen	4200	IT
7	109	Faviet	9000	Finance
8	110	Chen	8200	Finance

INNER JOIN :

Query3:

```
SELECT e.employee_id, e.last_name, j.job_title, d.department_name, l.city
FROM hr.employees e
JOIN hr.departments d ON e.department_id = d.department_id
JOIN hr.jobs j ON e.job_id = j.job_id
JOIN hr.locations l ON d.location_id = l.location_id
ORDER BY e.employee_id
FETCH FIRST 50 ROWS ONLY;
```

Output3:

	EMPLOYEE_ID	LAST_NAME	JOB_TITLE	DEPARTMENT_NAME	CITY
1	100	King	President	Executive	Seattle
2	101	Yang	Administration Vice President	Executive	Seattle
3	102	Garcia	Administration Vice President	Executive	Seattle
4	103	James	Programmer	IT	Southlake
5	104	Miller	Programmer	IT	Southlake
6	105	Williams	Programmer	IT	Southlake
7	106	Jackson	Programmer	IT	Southlake
8	107	Nguyen	Programmer	IT	Southlake

LEFT OUTER JOIN:

Query4:

```
SELECT e.employee_id, e.last_name, e.department_id, d.department_name
FROM hr.employees e
LEFT OUTER JOIN hr.departments d
ON e.department_id = d.department_id;
```

Output4:

	EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
1	200	Whalen	10	Administration
2	201	Martinez	20	Marketing
3	202	Davis	20	Marketing
4	114	Li	30	Purchasing
5	115	Khoo	30	Purchasing
6	116	Baida	30	Purchasing
7	117	Tobias	30	Purchasing
8	118	Himuro	30	Purchasing

RIGHT OUTER JOIN:

Query5:

```
SELECT e.employee_id, e.last_name, d.department_id, d.department_name
FROM hr.employees e
RIGHT OUTER JOIN hr.departments d
ON e.department_id = d.department_id;
```

Output5:

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	EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
1	200	Whalen	10	Administration
2	201	Martinez	20	Marketing
3	202	Davis	20	Marketing
4	114	Li	30	Purchasing
5	115	Khoo	30	Purchasing
6	116	Baida	30	Purchasing
7	117	Tobias	30	Purchasing
8	118	Himuro	30	Purchasing

FULL OUTER JOIN:

Query6:

```
SELECT e.employee_id, e.last_name, d.department_id, d.department_name
FROM hr.employees e
FULL OUTER JOIN hr.departments d
ON e.department_id = d.department_id;
```

Output6:

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	EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
1	100	King	90	Executive
2	101	Yang	90	Executive
3	102	Garcia	90	Executive
4	103	James	60	IT
5	104	Miller	60	IT
6	105	Williams	60	IT
7	106	Jackson	60	IT
8	107	Nguyen	60	IT